

Background & Intent

Why This Project?

With millions of books available, finding the right next read can be challenging, especially when a reader wants something similar to a book they already enjoyed. We chose to explore whether patterns in readers’ past ratings could help identify books they are likely to enjoy next, rather than relying simply on popularity or broad genre categories. Our project combines Goodreads user-book interactions with detailed book metadata to identify relationships between books and create ranked recommendations based on shared reader preferences. With approximately 229 million user-book interactions, our dataset provides a large-scale view of how readers’ preferences overlap across books.

Our Goal

Existing work demonstrates the potential of data-driven approaches to book recommendation. Devika et al. (2021) explored recommender-system approaches for book discovery, demonstrating how recommendation techniques can be applied to help readers identify relevant books. Haberlah (2025) took a more personalized approach by combining Goodreads reading history with OpenLibrary metadata to predict individual book preferences. His work highlights how reader ratings, book characteristics, and metadata can work together to better understand what a reader may enjoy.

Building on these approaches, our project focuses on a slightly different question: rather than predicting whether an individual reader will like a specific book, we ask whether the collective behavior of readers who liked a book can help identify what to recommend next. We use shared reader behavior and book-to-book similarity to rank potential recommendations, combining information about which readers liked the same books with characteristics of the books themselves. Ultimately, we want to answer: **“If you liked this book, what are you most likely to enjoy next?”**

Data Sources

goodreads

Goodreads logo. From Goodreads (2025).



From University of California, San Diego, by University of California, San Diego, 2026, Wikipedia

UCSD Goodread Interactions

PRIMARY

Collected by UCSD researchers from public Goodreads shelves in 2017. The CSV contains approximately 229 million interactions (~4.1 GB), with user ID, book ID, and rating as our key variables. Source link: <https://cseweb.ucsd.edu/~jmcauley/datasets/goodreads.html>.



Open Library logo. Open Library, n.d.

UCSD Goodreads Book Metadata

A downloadable dataset of approximately 2.3 million books, containing additional book-level metadata. The data are provided as a downloadable file and cover information through 2017. Source link: <https://cseweb.ucsd.edu/~jmcauley/datasets/goodreads.html>.

SECONDARY

OpenLibrary

Provides book-level bibliographic metadata, including title, author, publication date, subjects, ISBNs, publishers, languages, and ratings. Because of instability with the Open Library API, we used Open Library data dumps instead of querying the API directly. We use these metadata dumps to supplement Goodreads metadata and connect records using Goodreads ID.

Data Cleaning & Manipulation

	UCSD Interactions	UCSD Book Metadata	OpenLibrary
CREATE	• Sample 50,000 readers • Keep all sampled reader interactions • Map editions → work_id	• Create work-level book dimension • Standardize English codes to eng • Aggregate ratings/reviews	• Search by ISBN, title, and/or author • Build Goodreads → OpenLibrary matches
UPDATE	• Collapse duplicate reader–work records • Keep highest rating • is_read = 1 if either edition was read • Create 80/10/10 reader splits	• Collapse editions into work_id • Select most-rated edition as canonical • Recompute aggregate ratings	• Match canonical editions to OpenLibrary works • Aggregate ratings by work • Add subjects and publication metadata
DELETE	• Drop records with no matching work_id • Retain rating 0 as “not rated”	• Drop unused/high-cost columns • Remove duplicate edition-level records	• Keep only matching canonical Goodreads editions
FINAL	• 12.8M reader × work records • user_id, work_id, is_read, rating interactions_sample_50k.pkl	• 1.52M works • 34 columns book_dimensions.pkl	• 399K matched works • Subjects, ratings, ISBNs, dates, and work/edition IDs open_library_match.pkl

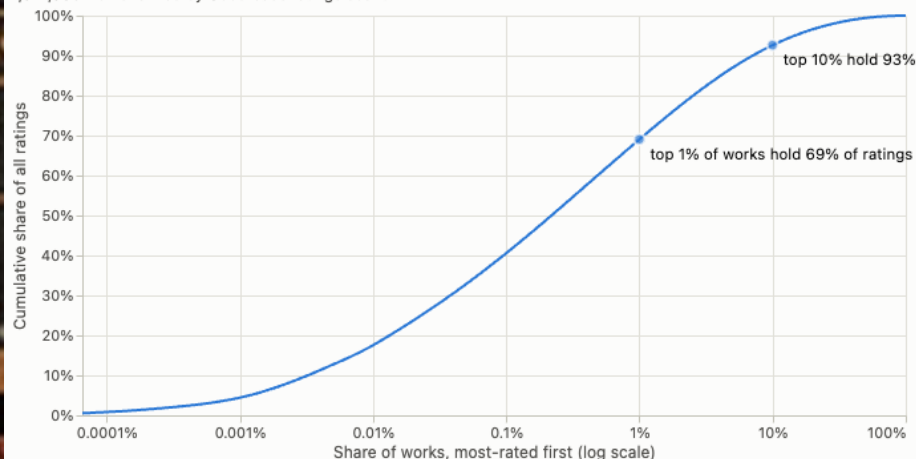
All major data manipulation steps are documented here. Other minor manipulations are noted in the notebooks.

Interactions Analysis

Our exploratory analysis revealed that ratings were highly concentrated among a relatively small number of works. Although our dataset contained 1,521,963 works, the top 1% of works accounted for 69% of all Goodreads ratings, while the top 10% accounted for 93%. This uneven distribution indicates that many books have very limited reader activity, making their similarity scores less reliable. To reduce the influence of sparsely rated books, we grouped positive interactions by book and established a minimum support threshold. We selected an initial threshold of 10 positive ratings per book, ensuring that each retained book had support from at least 10 readers before being considered by the recommender. This approach helps balance coverage with reliability and provides a stronger foundation for our Jaccard similarity and directional recommendation analyses, where comparisons based on very few readers could otherwise produce unstable or misleading results.

A small share of works holds most of the ratings

1,521,963 works ranked by Goodreads ratings count



Readers	All rows	Liked rows (4–5★)	Works with 10+ likes	Benchmark books usable	Build time
10,000	2.6M	0.83M	11,987	8,549	0.1 min
25,000	6.5M	2.08M	28,631	23,470	0.4 min
50,000	13.0M	4.12M	52,356	46,475	1.1 min
100,000	25.7M	8.27M	91,847	85,273	2.3 min

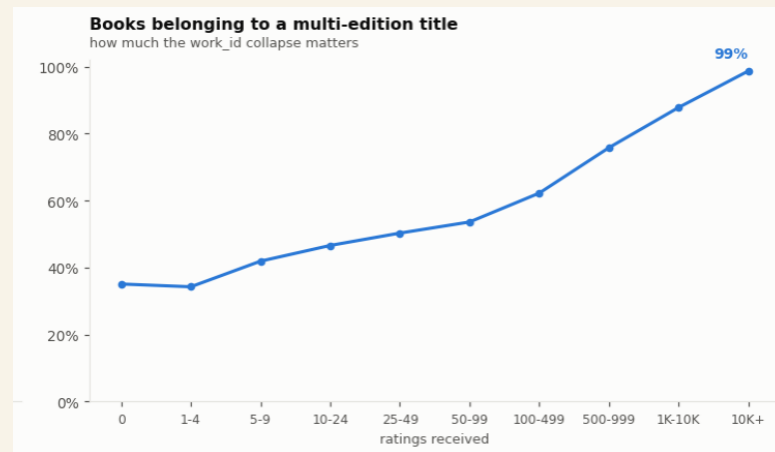
The interactions file contains 228.6 million rows, making it too large to fit into memory on a typical laptop or Google Colab environment. The cleaning notebook established a sampling approach that selects readers and retains all interactions associated with each sampled reader. Sampling individual rows instead would break the connections between books in a reader’s history, which are essential for a co-reading recommendation system. To determine an appropriate sample size, we evaluated five reader sample sizes across three criteria: the number of works with sufficient data to support recommendations, the number of Goodreads “readers also enjoyed” pairs that remained in the sample, and the time required to build the item-item model.

Book Metadata Analysis

We next examined the impact of multiple editions of the same work on our book-level data. For example, *Killing Floor* appears 50 times in the raw dataset across mass-market, hardcover, audiobook, Audio CD, Dwarsligger, and translated editions. If each edition remained a separate book, readers of different editions would not appear to share the same title, and the recommender could incorrectly prioritize other editions of the same work. We therefore consolidated editions using Goodreads’ existing `work_id`, with the most-rated edition selected as the canonical record and its descriptive fields retained. We summed `ratings_count` and `text_reviews_count` across editions and recomputed `avg_rating` as a ratings-weighted mean, while retaining capped fields such as `n_shelves` and `n_similar` rather than summing them. We also standardized English language codes to `eng` and removed unused or redundant columns to reduce the size and complexity of the metadata. This work-level consolidation ensured that reader activity and book characteristics were represented at the level of the actual work rather than individual editions, providing a more meaningful foundation for calculating book similarity and recommendations. Overall, the visual analysis showed that collapsing editions to the work level was important for preventing duplicate representations of the same story and ensuring that reader activity was accurately combined before calculating book similarity and recommendations.

When book editions are rolled up into the works, we see a see the changes that are featured below in the table.

	source (editions)	book dimension (works)	note
measure			
rows	2,360,655	1,521,963	35.5% fewer rows
columns	29 (+10 genre weights)	34	7 dropped, lists kept, genres flattened
ratings counted	958,938,616	958,938,616	preserved exactly
in memory	2.34 GB	1.49 GB	



Backend Similarity Scores

Rather than relying on a single similarity model, we tried to combine the rankings from Jaccard, Directional Similarity, and “Finished Together” (another Jaccard score) using Ranked Fusion. Each model contributes a score based on where a book appears in its ranked list, so higher-ranked recommendations receive more weight while still allowing agreement across models to matter. Books that are highly ranked by multiple models receive a stronger combined score, allowing the final recommendations to balance different signals of reader similarity. This creates a single, deterministic ranking that reflects agreement across all three approaches rather than relying on any one similarity measure. Here is an explanation of the 3 similarity scores we explored and the question they answer.

Jaccard Similarity

Jaccard scoring asks **“How much do the readers who liked Book A overlap with the readers who liked Book B?”**

For each book, we create a set of readers who gave it a 4 or 5 star rating. Then we compare those two sets. A higher Jaccard score means the two books have a greater overlap in their audiences. Jaccard's denominator includes everyone who liked either book, so it can be affected by book popularity.

“Finished Together”

Finished together asks, **“How much do the books that readers actually finished overlap?”**

We compare the set of other books that readers marked as read and calculate their Jaccard similarity based on this co-presence. This approach uses reading behavior rather than ratings, capturing books that readers actually completed instead of books they only intended to read. A higher score indicates greater overlap in the books readers finished together.

Directional Recommendation

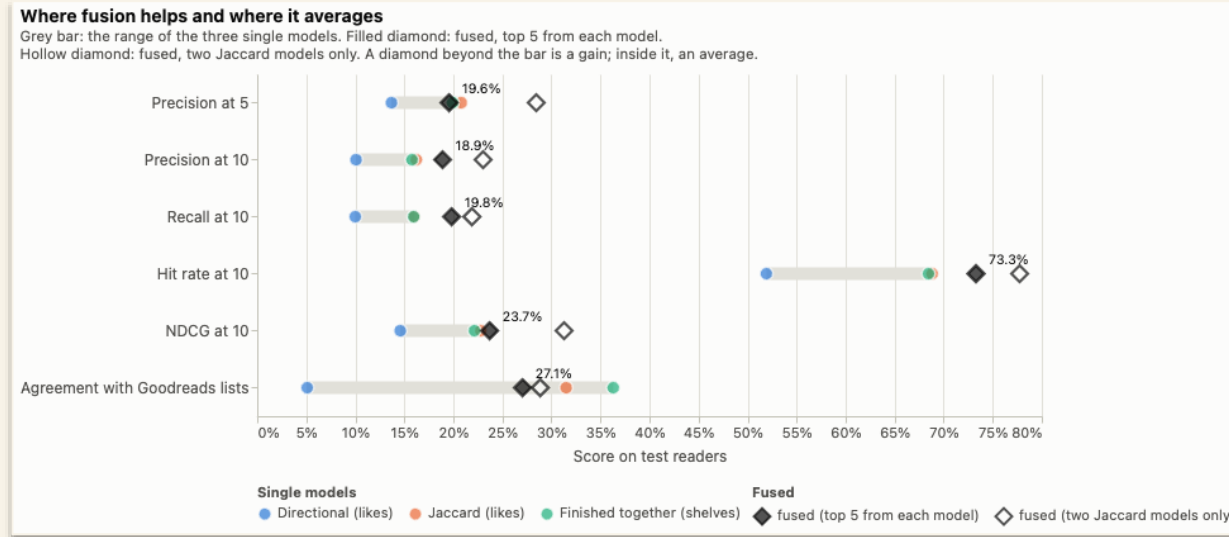
Directional scoring asks **“Of the people who liked Book A, what percentage also liked Book B?”**

We look at readers who gave Book A a 4 or 5 star rating and calculate the percentage of those readers who also rated Book B 4 or 5 stars. A higher directional score means Book B was more frequently liked by readers who liked Book A. This score is directional, so the relationship from Book A to Book B can differ from Book B to Book A.



Final Ranked Fusion Recommendations

The rank-fusion results show that combining models can improve recommendation performance, but Directional Similarity behaves differently from the two Jaccard-based approaches. While the three-model fusion performed well on several metrics, Directional Similarity was less aligned with the other similarity signals. The two-Jaccard fusion produced comparable or better performance across most evaluation metrics while providing a more consistent measure of reader overlap. Based on these results, we decided to remove Directional Similarity from the final recommender and use rank fusion between Jaccard Similarity and “Finished Together”. This allows the final recommendations to focus on two complementary measures of reader overlap and shared reading behavior without the directional model pulling the results toward a substantially different recommendation pattern.

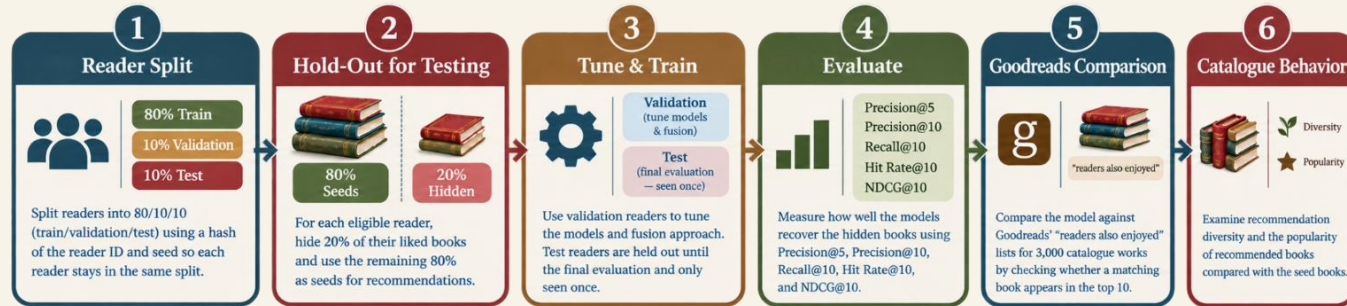


The visualization above shows how Directional Similarity performed compared to the two Jaccard methods. The diamonds represent how Directional Similarity affects the overall fused score.



Evaluation of Model

We evaluate the system using held-out readers from an 80/10/10 train, validation, and test split. For each eligible reader, 20% of their liked books are hidden, while the remaining 80% serve as seeds for generating recommendations. The validation readers are used to tune the models and fusion approach, while the test readers are held out until the final evaluation and are only seen once. The models are evaluated on how well they recover the hidden books using Precision@5, Precision@10, Recall@10, Hit Rate@10, and NDCG@10. We also compare the model against Goodreads’ “readers also enjoyed” lists for 3,000 catalogue works by checking whether a matching book appears in the model’s top 10. Finally, we examine catalogue behavior by measuring recommendation diversity and the popularity of recommended books compared with the seed books.



Ethics of Book Recommendations

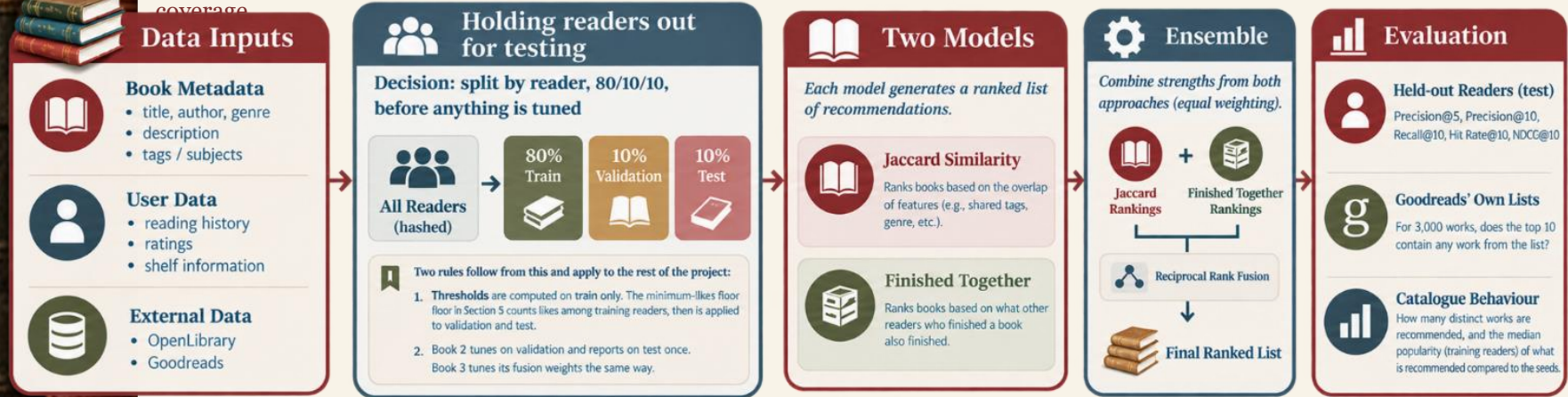
Book recommendation systems can unintentionally reinforce existing biases in readership and publishing, potentially limiting exposure to diverse authors and perspectives. Our Goodreads data also comes from publicly available user activity that was scraped without clear user consent, raising privacy and data-use concerns. In addition, recommendations may surface books with offensive, harmful, or controversial content without accounting for a reader’s individual boundaries.

In future iterations, we could mitigate these risks by clearly communicating our data sources and how recommendations are generated, auditing recommendations for author, genre, and perspective diversity, and providing users with more context about recommended books and potentially sensitive content. We could also explore ways to give readers greater control over the types of books included in their recommendations.



Recommendation Output

The recommendation system uses a reader-level 80/10/10 train, validation, and test split to help prevent data leakage and ensure recommendations are generated for readers the models have not seen. Jaccard Similarity and Finished Together provide complementary recommendation approaches, which are combined through equally weighted reciprocal rank fusion to create a final ranked list. The system produces ranked, book-level recommendations for each input book, allowing the strongest recommendations to be surfaced first and presented as a personalized set of potential reads. Overall, the outputs demonstrate how the system combines different reader-behavior signals to generate recommendations while supporting relevance, similarity, and broader catalogue coverage.



Gone Girl recommendation		The Hobbit recommendation		The Night Circus recommendation	
1	The Fault in Our Stars	1	Harry Potter and the Sorcerer's Stone (Harry P...	1	The Ocean at the End of the Lane
2	The Girl on the Train	2	Harry Potter and the Prisoner of Azkaban (Harr...	2	Miss Peregrine's Home for Peculiar Children (M...
3	The Help	3	Harry Potter and the Goblet of Fire (Harry Pot...	3	Water for Elephants
4	The Girl with the Dragon Tattoo (Millennium, #1)	4	A Game of Thrones (A Song of Ice and Fire, #1)	4	The Book Thief
5	The Hunger Games (The Hunger Games, #1)	5	Harry Potter and the Deathly Hallows (Harry Po...	5	The Hunger Games (The Hunger Games, #1)

Next Steps & Takeaways

1

An ML layer can be added to improve recommendations.

A machine learning layer could learn from user behavior and recommendation outcomes to improve the accuracy of recommendations. It could identify patterns that our current similarity-based approach may not capture. This would allow the system to become more personalized as more reader data become available.

2

Book covers can be pulled from OpenLibrary for more interesting visuals.

Adding book covers from Open Library would make the recommendation interface more visually engaging. Covers would also give users an immediate way to recognize and distinguish recommended books. This could make the dashboard feel more like a real-world book discovery tool.

3

Real world AB test to see which book recommendation actually have an effect to the reader.

A real-world A/B test could compare different recommendation methods with actual readers. We could measure whether one approach leads readers to select, read, or rate recommended books more frequently. This would provide evidence of whether our recommendations are actually influencing reader behavior rather than relying only on similarity scores. This type of testing could also influence the weights we give the different methods in the end.

Similarity scores provide useful recommendations, but they don't tell the whole story.

Jaccard similarity and our directional recommendation score helped quantify relationships between books based on shared readers. However, popularity and data availability can influence these scores, which is why we explored thresholds, metadata, and multiple measures rather than relying on a single metric. Our prototype demonstrates a useful approach, but machine learning, richer visuals, and real-world A/B testing could make recommendations more accurate and measurable.



References

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Statement of Work

We worked well together as a group and met regularly to keep the project moving forward. In the future, using a Kanban board or another project management tool could help make deadlines, responsibilities, and progress clearer, ultimately making our workflow more organized and efficient.

Olivia Cain

David Hook

Sarah Ehlen

- Contributed to the project proposal
- Interactions cleaning & EDA
- Developed the directional recommendation and Jaccard similarity methods

- Ranked fusion & Finished Together models
- Python module breakout & notebook consolidation
- Book metadata cleaning

- Project proposal
- OpenLibrary API EDA & cleaning
- Assisted in notebook merging
- Created project report and edited on report based on comments